

HYDROGENATED NBR (HNBR) SYNTHESIS AND LOW-MOLECULAR-WEIGHT PROCESS PATENTS

1. ABSTRACT

This report synthesizes patent evidence relevant to hydrogenated nitrile butadiene rubber (HNBR), with emphasis on how NBR precursors are transformed into hydrogenated or partially hydrogenated products and how process conditions are tuned to control residual unsaturation, molecular weight, and catalyst residues. Across the selected patents, the target material is consistently defined as a nitrile rubber derived from at least one conjugated diene and at least one α,β -unsaturated nitrile, with hydrogenation of the diene-derived double bonds to improve heat, oil, and ozone resistance. Several patents focus on lowering Mooney viscosity and narrowing molecular weight distribution by combining metathesis-driven chain scission with subsequent hydrogenation, while others address catalyst selection and post-treatment to improve ageing stability or reduce residual noble metals.

The landscape is dominated by two major synthesis strategies. First, metathesis-assisted molecular-weight reduction is disclosed in multiple patents, typically using ruthenium-based Grubbs/Hoveyda-type catalysts, sometimes with co-olefins, followed by hydrogenation to yield low-MW HNBR. These routes emphasize simultaneous or sequential depolymerization and hydrogenation, with reported temperatures commonly in the 60–200 °C range and hydrogen pressures from 700,000 to 15,000,000 Pa. Second, catalyst-management patents focus on improving HNBR quality after hydrogenation by minimizing or removing Ru, Rh, or Pd residues using filtration, adsorption, ion exchange, membranes, or ionic liquids. A separate formulation patent addresses guanidine-free curing systems for HNBR compounds, reflecting downstream processing needs rather than synthesis.

Overall, the evidence indicates that the most effective HNBR synthesis routes combine controlled metathesis degradation to reduce feedstock viscosity, followed by hydrogenation under moderate-to-high pressure with ruthenium, palladium, or rhodium catalysts. The most important process trends are catalyst selectivity, low residual double bond content, retention of polymer integrity, and practical work-up methods that preserve ageing performance and reduce discoloration.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis catalyst systems for reacting nitrile rubber in the presence of a salt additive, with the stated goal of improving metathesis activity. The process is relevant to HNBR because the patent explicitly discusses metathesis degradation of nitrile rubber as a precursor step before hydrogenation to low-molecular-weight HNBR. - Salt: LiBr — Example 7 - Salt: CsBr — Example 8 - Salt amount: 1 mg — Example 7 - Salt amount: 1 mg — Example 8 - Grubbs II catalyst amount: 10 mg — Example 7 - Grubbs II catalyst amount: 10 mg — Example 8 - Chlorobenzene: 0.3 ml — Example 7 - CDCl₃: 0.2 ml — Example 7 - Diethyl diallylmalonate addition time: 2 minutes — Example 7 - Diethyl diallylmalonate addition time: 2 minutes — Example 8 - Reaction temperature: room temperature — Example 7 - Reaction temperature: room temperature — Example 8 - Salt:catalyst weight ratio: 0.01:1 to 10000:1 — Claims

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the salt-free control. - Example 8: CsBr also affected ring-closing metathesis, but with a different conversion profile than LiBr.

Technical Relevance Although not an HNBR hydrogenation patent per se, it is relevant because it teaches salt-accelerated metathesis chemistry on nitrile-rubber-related substrates. Such catalyst activation concepts can support the chain-scission step used to prepare low-molecular-weight HNBR precursors.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses production of low-molecular-weight hydrogenated nitrile rubber by first dissolving rubber in monochlorobenzene, charging the cement to a reactor, degassing with ethylene or nitrogen, and then carrying out metathesis/hydrogenation-related processing at elevated temperature. The examples show that catalyst loading and co-olefin choice are used to control molecular weight reduction and final Mooney viscosity. - Rubber: 200 g — Example 1 - Monochlorobenzene: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber: 200 g — Example 2 - Monochlorobenzene: 1133 g — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber: 75 g — Example 3 - Monochlorobenzene: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Rubber: 75 g — Example 4 - Monochlorobenzene: 1175 g — Example 4 - Solid content: 6 wt.-% — Example 4 - 1-hexene: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity: 73 — starting polymer, Table 2 - Mooney viscosity: 43 — Experiment 1, Table 2 - Mooney viscosity: 28 — Experiment 2, Table 2 - Mooney viscosity: 3 — Experiment 3, Table 2 - Mooney viscosity: 31 — Experiment 4, Table 2 - Mn: 85 kg/mol — substrate, text - Mw: 296 kg/mol — substrate, text - Mn: 27 kg/mol — typical product, text - Mw: 54 kg/mol — typical product, text - PDI: 3.4 — substrate, text - PDI: 2.0 — typical product, text

Relevant Experimental Evidence - Example 1: ethylene-degassed cement at 15 wt.-% solids was used as the starting point for molecular-weight reduction. - Example 2: the same 15 wt.-% cement and ethylene pressure were used to compare catalyst performance. - Example 3: lower-solids cement (6 wt.-%) was used to further reduce molecular weight and Mooney viscosity. - Example 4: 1-hexene was used as co-olefin, demonstrating an alternative metathesis feed strategy.

Technical Relevance This is directly relevant because it teaches a route to low-MW HNBR via metathesis-based degradation followed by hydrogenation. The patent provides concrete process windows and property outcomes that are central to HNBR precursor design.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by metathesis of NBR in solution using co-olefins and ruthenium metathesis catalysts, followed by hydrogenation. The process is designed to reduce molecular weight and narrow the molecular weight distribution while retaining HNBR functionality. - Rubber: 200 g — Example 1 - Monochlorobenzene: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber: 200 g — Example 2 - Monochlorobenzene: 1133 g — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber: 75 g — Example 3 - Monochlorobenzene: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Rubber: 75 g — Example 4 - Monochlorobenzene: 1175 g — Example 4 - Solid content: 6 wt.-% — Example 4 - 1-hexene: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity: 73 — starting polymer, Table 2 - Mooney viscosity: 43 — Experiment 1, Table 2 - Mooney viscosity: 28 — Experiment 2, Table 2 - Mooney viscosity: 3 — Experiment 3, Table 2 - Mooney viscosity: 31 — Experiment 4, Table 2 - Mn: 85 kg/mol — substrate, text - Mw: 296 kg/mol — substrate, text - Mn: 27 kg/mol — typical product, text - Mw: 54 kg/mol — typical product, text - PDI: 3.4 — substrate, text - PDI: 2.0 — typical product, text

Relevant Experimental Evidence - Example 1: ethylene at 6.9×10^5 Pa was used to drive metathesis of a 15 wt.-% rubber cement. - Example 2: the same conditions were repeated to compare reproducibility. - Example 3: lower solids content gave a much lower Mooney viscosity product. - Example 4: 1-hexene co-olefin was used and still produced a low-MW hydrogenated product.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR, a key objective in HNBR synthesis. It demonstrates how metathesis conditions and co-olefin choice influence final viscosity and molecular-weight distribution.

Patent 4

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubbers with controlled calcium and chlorine contents and specific thio end groups, together with improved coagulation and washing conditions. It is not a hydrogenation patent, but it is relevant as a precursor-material patent because it addresses NBR feedstock quality and work-up, which

influence downstream HNBR hydrogenation behavior. - Acrylonitrile content: 38.6% by weight — Example 36 - Latex concentration: 10.9 — Example 36 table - Salt: CaCl₂ — Example 36 - Salt concentration: 0.3% by weight — Example 36 table - Salt based on NBR: 1.8% by weight — Example 36 table - Coagulation temperature: 90 °C — Example 36 table - Washing temperature: 60 °C — Example 36 table - Washing time: 8 h — Example 36 table - Chlorine content: 100 ppm — Example 36 table - Calcium content: 1930 ppm — Example 36 table - MV1: 47 — Example 36 table - MV2: 51 — Example 36 table - Storage stability: 4 — Example 36 table

Relevant Experimental Evidence - Example 36: CaCl₂ coagulation and calcium-ion washing produced an NBR with defined chlorine and calcium contents and measured storage stability.

Technical Relevance The patent matters for HNBR synthesis because the quality of the NBR precursor strongly affects hydrogenation, work-up, and final rubber performance. It highlights how coagulation and residual-ion control can be engineered before hydrogenation.

Patent 5

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions of nitrile rubber, using ruthenium- or osmium-based metathesis catalysts together with salt additives such as alkaline earth metal chlorides. The process is aimed at improving catalyst activity for nitrile-rubber degradation, which is a precursor step to low-molecular-weight HNBR production. - Degree of hydrogenation: 50 to 100% — background discussion - Mooney viscosity: 55 to 120 — commercially available HNBR, background - Residual double bond content: 1 to 18% — background - Mooney viscosity threshold: below 55 — problem statement - Mn threshold: <200000 g/mol — problem statement - Mooney viscosity of NBR feedstock: about 30 Mooney units — background - Mn feedstock threshold: <70000 g/mol — problem statement - Degradation temperature: 30 to 100°C — preferred embodiment - Hydrogenation temperature: 60 to 150°C — preferred embodiment - Salt additive: alkaline earth metal chloride — Claims - Salt additive: calcium chloride or magnesium chloride — Claims

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it improves the metathesis step used to lower NBR molecular weight before hydrogenation. Better catalyst activity and salt-assisted metathesis can translate into more efficient HNBR precursor preparation.

Patent 6

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of at least partially unsaturated NBR in solution using ruthenium, palladium, or rhodium compounds, with the explicit goal of producing HNBR having improved ageing properties without a separate noble-metal removal step. The process emphasizes catalyst loading control, hydrogen pressure, temperature, and reaction time to balance conversion and product stability. - Conjugated diene content: 40% to 90% by weight — Claims - α,β -unsaturated nitrile content: 10% to 60% by weight — Claims - Ruthenium amount: 10 ppm to 200 ppm — Claims - Ruthenium amount: 10 ppm to 150 ppm — Claims - Ruthenium amount: 10 ppm to 120 ppm — Claims - Ruthenium amount: 10 ppm to 79 ppm — Claims - Ruthenium amount: 42 ppm to 79 ppm — Claims - Palladium amount: 20 ppm to <67 ppm — Claims - Palladium amount: 44 ppm to <67 ppm — Claims - Rhodium amount: 50 ppm to <270 ppm — Claims - Rhodium amount: 79 ppm to <270 ppm — Claims - Rhodium amount: 50 ppm to 250 ppm — Claims - Rhodium amount: 50 ppm to 170 ppm — Claims - Rhodium amount: 79 ppm to 170 ppm — Claims - Hydrogenation temperature: 60 to 200°C — Claims - Hydrogenation pressure: 700000 to 15000000 pascals — Claims - Hydrogenation time: 1 to 24 hours — Claims - Rhodium content in HNBR: 58-60 ppm — prior art discussion - Rhodium content in purified HNBR: 5.8 to 9.2 ppm — prior art discussion - Ruthenium content: very low — prior art discussion - Hydrogenation temperature: 140°C — ageing test discussion - Ageing time: 4 days — ageing test discussion - Mooney viscosity increase limit: less than 40% — ageing test discussion - Gel content increase limit: less than 10% — ageing test discussion

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This is one of the most directly relevant patents because it defines a practical HNBR hydrogenation window with explicit catalyst loading and operating conditions. It also addresses a major industrial issue for HNBR: residual noble-metal contamination and ageing stability.

Patent 7

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed

- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This European patent mirrors the US 2021 filing and discloses solution hydrogenation of at least partially unsaturated NBR using Ru, Pd, or Rh compounds to produce HNBR with improved ageing properties. The emphasis is on maintaining product quality while avoiding elaborate post-hydrogenation noble-metal removal steps. - Conjugated diene content: 40% to 90% by weight — Claims - α,β -unsaturated nitrile content: 10% to 60% by weight — Claims - Ruthenium amount: 10 ppm to 200 ppm — Claims - Ruthenium amount: 10 ppm to 150 ppm — Claims - Ruthenium amount: 10 ppm to 120 ppm — Claims - Ruthenium amount: 10 ppm to 79 ppm — Claims - Ruthenium amount: 42 ppm to 79 ppm — Claims - Palladium amount: 20 ppm to <67 ppm — Claims - Palladium amount: 44 ppm to <67 ppm — Claims - Rhodium amount: 50 ppm to <270 ppm — Claims - Rhodium amount: 79 ppm to <270 ppm — Claims - Rhodium amount: 50 ppm to 250 ppm — Claims - Rhodium amount: 50 ppm to 170 ppm — Claims - Rhodium amount: 79 ppm to 170 ppm — Claims - Hydrogenation temperature: 60 to 200°C — Claims - Hydrogenation pressure: 700000 to 15000000 pascals — Claims - Hydrogenation time: 1 to 24 hours — Claims - Rhodium content in HNBR: 58-60 ppm — prior art discussion - Rhodium content in purified HNBR: 5.8 to 9.2 ppm — prior art discussion - Ruthenium content: very low — prior art discussion - Hydrogenation temperature: 140°C — ageing test discussion - Ageing time: 4 days — ageing test discussion - Mooney viscosity increase limit: less than 40% — ageing test discussion - Gel content increase limit: less than 10% — ageing test discussion

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is highly relevant because it provides a modern industrial hydrogenation window for HNBR and explicitly links catalyst residue control to ageing performance. It is useful for selecting practical hydrogenation conditions and catalyst loadings.

Patent 8

- Patent Number: EP2292687B1
- Patent Title: ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses guanidine-free vulcanisable mixtures containing HNBR-based rubbers together with non-aromatic amine or amidine accelerators and retarder systems such as benzoic acid or carbodiimides. It is a downstream formulation patent rather than a synthesis patent, but it is relevant because it shows how HNBR is compounded and cured in practical applications. - HNBR1 residual double bond content: 0.4% — Example 2 - HNBR1 maleic monoethylester content: 4.5% — Example 2 - HNBR1 Mooney viscosity: 101 — Example 2 - NBR1 butadiene content: 60.4% — Table A - NBR1 termonomer: maleic acid monoethyl ester — Table A - NBR1 termonomer content: 4.5% — Table A - NBR1 acrylonitrile content: 35.1% — Table A - NBR1 Mooney viscosity: 35 — Table A - AEM/ACM/HNBR accelerator amount: 0.1 to 10 wt.% — synthesis passage - Preferred accelerator amount: 0.5 to 8 wt.% — synthesis passage - Most preferred accelerator amount: 1.0 to 5.0 wt.% — synthesis passage - Mixing speed: 40 RPM — Example 1 - Filling factor: 70% — Example 1 - Dump temperature: 100 °C — Example 1 - Open mill friction: 1:1.25 — Example 1 - Open mill temperature: 40 °C — Example 1 - Mixing time: 6 min — Example 1 - Press-curing temperature: 180 °C — Example 1 - Press-curing pressure: 100 bar — Example 1 - Post-curing temperature: 175 °C — Example 1 - Post-curing time: 4 hours — Example 1 - AEM/ACM/HNBR polymer content: 100 phr — Example 1 and 2 tables - Carbon black content: 47.50 phr — Example 1 - Carbon black content: 40.0 phr — Example 2 - Stabaxol I content: 1.12 phr — Example 1 - Stabaxol I content: 1.0 or 1.5 phr — Example 2

Relevant Experimental Evidence - Example 1: compared guanidine-containing and guanidine-free cure systems for ACM-based compounds. - Example 2: showed HNBR-based compounds with Stabaxol I and different accelerator/retarder combinations, including a hydrogenated nitrile terpolymer with 0.4% residual double bonds.

Technical Relevance This patent is relevant to HNBR because it demonstrates practical curing packages for hydrogenated nitrile rubber compounds. It helps connect HNBR synthesis to downstream processing, scorch safety, and final elastomer performance.

Patent 9

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses novel ruthenium carbene complexes and their use in metathesis reactions, including depolymerization of NBR and subsequent hydrogenation to HNBR. The examples focus on synthesis of the Ru complexes themselves, while the broader disclosure emphasizes their use in metathesis/hydrogenation of

nitrile rubber at elevated temperature and pressure. - Ligand synthesis scale: 5.0 mmol — Examples 1-20 - Ru complex synthesis scale: 1.0 mmol — Examples 1-20 - Reaction temperature: 85°C — Example 1 - Pd(PPh₃)₄ amount: 2% — Example 1 - K₂CO₃ amount: 1.5 eq — Example 1 - DME: 10 mL — Example 1 - Deionized water: 3 mL — Example 1 - DCM: 15 mL — Example 1 - CuCl: 270 mg — Example 1 - CuCl: 2.5 mmol — Example 1 - CuCl: 2.5 eq — Example 1 - Ligand 3a amount: 250 mg — Example 1 - Ligand 3a amount: 1.2 mmol — Example 1 - Ligand 3a amount: 1.2 eq — Example 1 - Ru complex 1b amount: 860 mg — Example 1 - Ru complex 1b amount: 1.0 mmol — Example 1 - Ru complex 1a amount: 830 mg — Example 2 - Ru complex 1a amount: 1.0 mmol — Example 2 - CuCl: 270 mg — Example 2 - CuCl: 2.5 mmol — Example 2 - CuCl: 2.5 eq — Example 2 - DCM: 15 mL — Example 2 - Ligand 3b amount: 250 mg — Example 2 - Ligand 3b amount: 1.2 mmol — Example 2 - Ligand 3b amount: 1.2 eq — Example 2 - Reaction time: overnight — Example 1 - Reaction time: 30-60 min — Examples 1-20 (Ru complex formation) - Ligand preparation scale: 5.0 mmol — Examples 3-20 - Hydrogenation temperature: 60-150°C — synthesis passage - Depolymerization temperature: 30-100°C — synthesis passage - Hydrogen pressure: high pressure — synthesis passage

Relevant Experimental Evidence - Example 1: a Ru complex was synthesized in 86% yield from a 5.0 mmol ligand-scale reaction, then converted to a green Ru complex in 4% yield. - Example 2: an alternative Ru complex was prepared in 91% ligand yield and 29% Ru-complex yield. - Examples 3-20: multiple ligand/Ru-complex variants were synthesized, showing the breadth of the catalyst family.

Technical Relevance This patent is relevant because it provides a family of metathesis catalysts intended for NBR depolymerization and HNBR formation. It supports the catalyst-development side of HNBR synthesis, especially for simultaneous or sequential metathesis/hydrogenation routes.

Patent 10

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight hydrogenated nitrile rubber by reacting nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based metathesis catalyst, optionally with a co-olefin. The process is explicitly designed for simultaneous metathesis and hydrogenation, enabling molecular-weight reduction and hydrogenation in one operation. - Hydrogen pressure: presence of hydrogen — Claims - Co-olefin: optionally at least one co-olefin — Claims - Catalyst

metal: ruthenium or osmium — Claims - Reaction temperature: 60 to 150°C — synthesis passage - Hydrogen pressure: high pressure — synthesis passage - Mooney viscosity target: 1 to 50 — synthesis passage - Mw target: 20,000 to 250,000 — synthesis passage - PDI target: less than 2.5 — synthesis passage - Conjugated diene type: C4-C6 — synthesis passage - Preferred diene: 1,3-butadiene — synthesis passage - Preferred diene: isoprene — synthesis passage - Preferred diene: 2,3-dimethylbutadiene — synthesis passage - Unsaturated nitrile: α,β -unsaturated nitrile — synthesis passage - Preferred unsaturated acids: fumaric acid, maleic acid, acrylic acid, methacrylic acid — synthesis passage - Grubbs 2nd generation catalyst: 1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene)(tricyclohexylphosphine)ruthenium(phenylmethylene) dichloride — prior art discussion - Hydrogenation catalyst examples: Wilkinson's catalyst ((PPh₃)₃RhCl) — prior art discussion - Mooney viscosity of HNBR from NBR feedstock: about 55 — prior art discussion - Mooney viscosity of NBR feedstock: about 30 — prior art discussion

Relevant Experimental Evidence - Example 1: HNBR 1, HNBR 2, and HNBR 3 were compared using different hydrogenation catalysts and co-catalyst conditions. - Example 1: HNBR 1 achieved the lowest RDB values over 240 min and the lowest Mooney viscosity (33.3). - Example 1: HNBR 2 and HNBR 3 showed higher Mooney viscosities (41.3 and 69.3), demonstrating catalyst dependence.

Technical Relevance This is one of the most important patents for HNBR synthesis because it directly teaches simultaneous metathesis and hydrogenation of NBR. It provides a route to low-MW HNBR with controlled residual unsaturation and improved processability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7772328B2 and EP2473281B1 both use metathesis-assisted molecular-weight reduction before or during hydrogenation, but EP2473281B1 explicitly claims simultaneous hydrogenation/metathesis in hydrogen, whereas US7772328B2 emphasizes sequential low-MW HNBR preparation with co-olefins.
- US7951875B2 and US7772328B2 disclose very similar solution-processing examples with monochlorobenzene, ethylene at 6.9×10^5 Pa, 80°C, and catalyst loadings of 0.05-0.25 phr; both show that lower solids and co-olefin choice strongly affect Mooney viscosity and Mw.
- US20210340285A1 and EP3827029A1 define essentially the same hydrogenation window: Ru/Pd/Rh catalyst loadings in the tens of ppm, 60-200°C, 700000-15000000 Pa, and 1-24 h, with the key difference being jurisdiction and publication format.
- US7737233B2 and US8957161B2 are catalyst-activation patents for nitrile-rubber metathesis; US7737233B2 focuses on alkaline earth metal chlorides, while US8957161B2 highlights bromide salts such as LiBr and CsBr.
- EP2519350B1 is the most catalyst-centric disclosure, providing a broad family of Ru complexes for metathesis/depolymerization, whereas the HNBR-focused patents mainly optimize process conditions and catalyst residues rather than catalyst structure.

- EP2292687B1 is downstream and formulation-oriented rather than synthetic: it uses HNBR in guanidine-free cure systems, while the other patents focus on making the HNBR precursor or hydrogenated polymer itself.
- US7923518B2 is the least directly hydrogenation-related; it addresses NBR precursor coagulation, calcium/chlorine content, and storage stability, which can still influence downstream HNBR manufacture.

4. CONCLUSION

The patent set shows that the most effective HNBR synthesis strategies combine controlled NBR depolymerization or metathesis with subsequent or simultaneous hydrogenation under carefully selected catalyst and pressure conditions. For low-MW HNBR, the strongest evidence supports solution metathesis using ruthenium-based catalysts, often with co-olefins and catalyst-activation salts, followed by hydrogenation at roughly 60-200°C and 700000-15000000 Pa. Catalyst loading in the tens of ppm for Ru/Pd/Rh systems is repeatedly associated with practical hydrogenation performance and improved ageing behavior. For precursor quality, NBR coagulation and residual-ion control remain important, while downstream compounding benefits from guanidine-free cure systems. Recommended synthesis parameters based on the evidence are: metathesis in solution with Ru catalysts and optional co-olefin, hydrogenation at 60-150°C when simultaneous processing is desired, or 60-200°C for conventional hydrogenation, with catalyst levels minimized but sufficient to maintain conversion and product stability, and with post-process attention to noble-metal residue control.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US7923518B2 | Nitrile rubbers | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7923518B2/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2292687B1 | ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ... | Not disclosed | EP | 2012 |

<https://patents.google.com/patent/EP2292687B1/en>

- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>